

# ML Practice

## Stock Market & Asset Valuation — Predicting Closing Prices

Dataset: `stock_market_data.csv`

Target Variable: `Close` (next day's closing price)

### Scenario:

You're a new analyst on a research team. You've been given a clean daily price history (Open, High, Low, Close, Volume) for a few stocks and cryptocurrencies. Your job: build your **first Machine Learning model** to predict the next day's closing price.

💡 The dataset is already clean — no missing values or errors. You'll spend your time on the most important part: **features and modeling**.

## Section 1 — Load & Explore the Dataset

**Goal:** Get familiar with the raw data before touching anything.

### Tasks:

1. Load `stock_market_data.csv` and parse `Date` as a datetime.
2. Show the first few rows and check the shape and data types.
3. List the unique tickers.
4. Plot the closing price over time for **one stock** and **one cryptocurrency**.

**Checkpoint:** In 1–2 sentences, how does the crypto price chart look different from the stock price chart?

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## Section 2 — Pick an Asset & Set Up Your Target

### Tasks:

1. Choose **one ticker** to work with (e.g., `AAPL`).
2. Filter the data to just that ticker and sort by `Date`.
3. Create a new column `Next_Close` = tomorrow's closing price (shift the `Close` column by one day).

4. Drop the last row, since it has no "tomorrow."

**Quick Prompt:** Why are we predicting *tomorrow's* price using *today's* data instead of today's price using today's data?

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## Section 3 — Create a Few Simple Features

**Goal:** You don't need dozens of features a few good ones are enough for your first model.

**Tasks:**

1. **Daily Return** — percent change in **Close** from the day before.
2. **5-Day Moving Average** — rolling average of **Close** over the last 5 days.
3. **Previous Close (Lag Feature)** — yesterday's **Close** price as its own column.
4. Drop any rows with missing values created by these rolling/lag calculations.

✓ **Checkpoint:** In one sentence each, why might these three features help predict tomorrow's price?

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## Section 4 — Train-Test Split

**Goal:** Prepare your data to fairly evaluate your model's performance.

**Tasks:**

1. Set **X** (features) = your engineered features from Section 3.
2. Set **y** (target) = **Next\_Close**.
3. Split into training and testing sets using the **first 80% of dates for training** and the **last 20% for testing** — do **not** shuffle randomly.

**Quick Prompt:** Why shouldn't we randomly shuffle dates when splitting time-series data like this?

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## Section 5 — Build Your First Machine Learning Model

**Tasks:**

1. Train a **LinearRegression** model on your training data.

2. Predict **Next\_Close** on your test data.
3. Create a simple table comparing **Actual vs. Predicted** prices.
4. Plot Actual vs. Predicted prices on the same chart.

✓ **Checkpoint:** Does your model roughly follow the real price trend?

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## Section 6 — Model Evaluation

Calculate:

- **MAE**
- **RMSE**
- **R<sup>2</sup> Score**

📝 **Reflection:**

1. What does your R<sup>2</sup> score tell you about your model?
  2. In plain terms, what does your MAE mean (e.g., "predictions are off by about \$X on average")?
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## 🏁 Wrap-Up

You just completed a full ML workflow:

**Load → Feature Engineer → Split → Train → Predict → Evaluate.**

This is the same process used in real financial forecasting — just applied here on a simplified beginner dataset.